

Hamdy Freeg

Giza, Egypt

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[LinkedIn Profile](#) — [GitHub Profile](#)

Professional Summary

Passionate Computer Science student with a solid mathematical background, focusing on Backend Development and Systems Engineering. I enjoy building core logic, understanding low-level network mechanics, and solving complex algorithmic problems. Actively seeking a Software Engineering internship to apply my technical skills, write clean code, and gain hands-on experience in a real-world production environment.

Education

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|---|-----------------|
| El Shorouk Academy | [2025] – [2028] |
| <i>B.Sc. in Computer Science</i> | Cairo, Egypt |
| <ul style="list-style-type: none">• Current Grade: Excellent• Focus Areas: Data Structures & Algorithms, Object-Oriented Programming, Systems Engineering. | |
| Cairo University | [2024] – [2025] |
| <i>Studies in Mathematics (Transferred)</i> | Giza, Egypt |
| <ul style="list-style-type: none">• Relevant Coursework: Linear Algebra, Calculus, Probability & Statistics, Mathematical Logic. | |

Courses & Certifications

- **C++ Data Structures, Algorithms & 4 Critical Skills Masterclasses** – Dr. Mostafa Saad (*Udemy*)
- **Computer Science Foundations:** CS 61A (UC Berkeley), Mathematics for CS (MIT), From Nand to Tetris.
- **Backend & Systems:** Java Development SE, OOP in Java (UCSD), Database Design in PostgreSQL (UMich), Networking.

Technical Skills

- **Programming Languages:** C++, Java, JavaScript, TypeScript, Bash Scripting, Python, SQL.
- **Core Concepts:** Backend Development, Systems Engineering, Data Structures & Algorithms, OOP, Networking, DBMS, PostgreSQL.
- **Tools & Environments:** Linux, Git, GitHub, VS Code, IntelliJ, LaTeX.

Projects

- **DNS Server (C++):** Developed a raw UDP DNS server matching RFC 1035 standards using POSIX sockets. Engineered a custom serialization and deserialization engine to process packet headers, questions, and answers, successfully resolving complex DNS pointer compression.
- **ASKI System (Ask.fm Clone):** Built a platform utilizing core OOP principles and custom relational logic to handle user management, anonymous questions, and chronological thread tracking with persistent data structures.
- **Library Management System:** Designed and implemented a robust management system featuring user authentication, multi-role access control (Admin/User), and efficient inventory tracking with strict focus on system correctness.
- **Custom Assembler (Java):** Developed a functional compiler toolchain component along with a formatting-agnostic file comparison tool to master low-level data flows and structural OOP patterns.

Languages

- **English:** Professional
- **Arabic:** Native